

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Seventh Semester B.Tech Degree Regular and Supplementary Examination December 2023 (2019 Scheme)

Course Code: CST415**Course Name: INTRODUCTION TO MOBILE COMPUTING****Max. Marks: 100****Duration: 3 Hours****PART A***Answer all questions, each carries 3 marks.*

Marks

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|----|--|-----|
| 1 | How does middleware and gateway functions in mobile computing application? | (3) |
| 2 | Why internet is called ubiquitous network? | (3) |
| 3 | Which are the different entities in GSM? | (3) |
| 4 | How a system could indicate the progress of a call to the caller? | (3) |
| 5 | What are the various strengths of SMS? | (3) |
| 6 | List out the limitations of GPRS. | (3) |
| 7 | Explain the different applications of Wireless LAN. | (3) |
| 8 | Compare 3G and WiFi. | (3) |
| 9 | Which are the types of attacks on static assets? | (3) |
| 10 | Write down the functional differences between 2G and 3G networks? | (3) |

PART B*Answer any one full question from each module, each carries 14 marks.***Module I**

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|----|---|----|
| 11 | a) Draw the three-tier architecture of Mobile computing. Describe the functions of these tiers. | 10 |
| | b) What are the limitations of mobile computing? | 4 |

OR

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| 12 | a) Explain in detail about the design consideration for mobile computing. | 10 |
| | b) Describe the functionalities of transaction processing middleware. | 4 |

Module II

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| 13 | a) What is multiple access? Describe FDMA and TDMA with the application areas and examples. | 8 |
| | b) What is handover? How is handover different from roaming? | 6 |

OR

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- 14 a) Differentiate LEO and GEO satellite. 6
b) What is GSM? In GSM network, there are some databases used for various purposes. Which are these databases? What are their functions? 8

Module III

- 15 a) How do you develop location based applications using SMS? 8
b) Give details about Quality Of Service parameters that GPRS supports. 6

OR

- 16 a) With a neat schematic explain the system architecture of GPRS 8
b) Write notes on billing and charging in GPRS. 6

Module IV

- 17 a) How could manage mobility in WLAN? 8
b) Differentiate between Ad hoc and Infrastructure mode. 6

OR

- 18 a) What is WLL? Explain the architecture of WLL with neat schematic. 8
b) List out the advantages of WLAN. 6

Module V

- 19 a) Explain in detail about the components of information security. 14

OR

- 20 a) What is MPLS? Explain the functionalities of MPLS. 8
b) Write short note on multimedia broadcast services 6
